

COURSE NAME & CODE: ECA 0303 – Signals and systems

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Experiment 7.2

Fourier Series of Half-Wave Rectifier Output

Aim:

To compute and analyze the Fourier series representation of the output of a half-wave rectifier. The Aim is to understand how the Fourier series can decompose a non-sinusoidal periodic signal into its constituent sinusoidal components (harmonics), and to observe how these harmonics contribute to the overall signal.

Program:

% Test Case: Fourier Series of Half-Wave Rectifier Output

clc;

clear;

close all;

% Define parameters

A = 1; % Amplitude

f = 1; % Frequency (Hz)

T = 1/f; % Time period

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dt = 0.01;      % Sampling interval
t = 0:dt:2*T;   % Time vector for two periods

% Input sinusoidal signal
x_t = A * sin(2*pi*f*t);

% Half-wave rectified output
y_t = max(x_t, 0);

% Number of harmonics
N_harmonics = 10;

% Fourier Series Coefficients
a0 = (2/T) * sum(y_t) * dt; % DC coefficient

an = zeros(1, N_harmonics);
bn = zeros(1, N_harmonics);

for n = 1:N_harmonics
    an(n) = (2/T) * sum(y_t .* cos(2*pi*n*f*t)) * dt;
    bn(n) = (2/T) * sum(y_t .* sin(2*pi*n*f*t)) * dt;
end

% Reconstruct the signal
y_reconstructed = (a0/2) * ones(size(t));

for n = 1:N_harmonics

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    y_reconstructed = y_reconstructed + ...
        an(n) * cos(2*pi*n*f*t) + ...
        bn(n) * sin(2*pi*n*f*t);
end

% Plot Original Half-Wave Rectified Signal
figure;

subplot(3,1,1);
plot(t, y_t, 'r', 'LineWidth', 1.5);
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Half-Wave Rectified Output Signal');

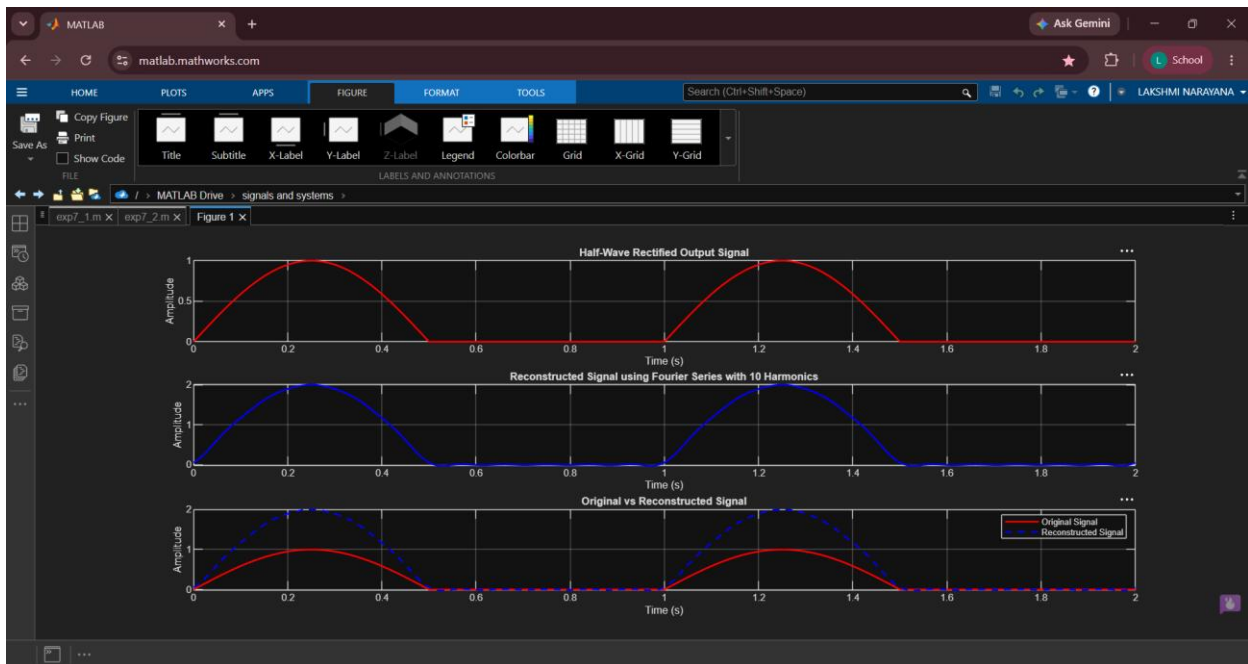
% Plot Reconstructed Signal
subplot(3,1,2);
plot(t, y_reconstructed, 'b', 'LineWidth', 1.5);
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title(['Reconstructed Signal using Fourier Series with ', ...
    num2str(N_harmonics), ' Harmonics']);

% Comparison Plot
subplot(3,1,3);
plot(t, y_t, 'r', 'LineWidth', 1.5);

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hold on;
plot(t, y_reconstructed, 'b--', 'LineWidth', 1.5);
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Original vs Reconstructed Signal');
legend('Original Signal', 'Reconstructed Signal');
```

Output:



Result:

- **Fourier Series Representation:** The Fourier series allows us to represent a non-sinusoidal periodic signal (like the half-wave rectified signal) as a sum of sinusoids. The DC component, fundamental frequency, and harmonics contribute to reconstructing the signal.

- **Effect of Harmonics:** The accuracy of the reconstruction improves as more harmonics are included in the Fourier series. However, a finite number of harmonics may introduce oscillations (Gibbs phenomenon) near discontinuities.
- **Practical Application:** The Fourier series is a powerful tool in signal processing, enabling the analysis and synthesis of signals, especially in systems where understanding the frequency components is critical.

This test case demonstrates the use of Fourier series to analyze and reconstruct the output of a half-wave rectifier, providing insights into the harmonic content and signal synthesis.